2020-B

1. $f(z)=\begin{cases} \dfrac{\bar{z}\,\mathrm{Re}(z)}{|z|^2} & z\neq 0 \\ 1 & z=0 \end{cases}$

$$\lim_{z\to 0}\frac{\bar{z}\,\mathrm{Re}(z)}{|z|^2}=\lim_{\substack{x\to 0\\ y\to 0}}\frac{x^2-ixy}{x^2+y^2}$$

令 $y=kx$

$$=\lim_{x\to 0}\frac{1-ik}{1+k^2}\neq 1 \quad \therefore 不连续$$

2. $w=z^2 \qquad xy=2$

$$w=z^2=(x+iy)^2=x^2-y^2+i2xy$$

$$\begin{cases} u=x^2-y^2 \\ v=2xy \end{cases}$$

$v=4 \quad (u\in R)$

3. $f(z)=x^3+3x^2yi-3xy^2-y^3i$

$u=x^3-3xy^2 \qquad u'_x=3x^2-3y^2 \qquad u'_y=-6xy$

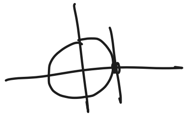
$v=3x^2y-y^3 \qquad v'_x=6xy \qquad v'_y=3x^2-3y^2$

$\therefore\ u'_x=v'_y=a \qquad u'_y=v'_x=-b$

$\therefore\ f'(z)=a+ib=(3x^2-3y^2)+i\,6xy$

$=3z^2$

4. $\displaystyle\int_{|z|=1} |z-1|\,|dz|$



$z = e^{i\theta} = \cos\theta + i\sin\theta$

$|dz| = |d\theta|$

$|z-1| = \sqrt{(\cos\theta-1)^2+\sin^2\theta}$

$= \sqrt{2-2\cos\theta}$

$$\int_0^{2\pi} \sqrt{4\left(\sin\frac{\theta}{2}\right)^2}\, d\theta$$

$$= 2\int_0^{2\pi} \sin\frac{\theta}{2}\, d\theta$$

$$= -4\cos\frac{\theta}{2}\Big|_0^{2\pi}$$

$$= -4(-1-1) = 8$$

5、6见2020-A

7. $v(x,y) = \dfrac{-2y}{(x+1)^2+y^2}$ 为调和 $\quad f(z) = u + iv$

$$v'_x = \frac{2y(2x+2)}{[(x+1)^2+y^2]^2} = b \qquad v''_{xx} + v''_{yy} = 0$$

$$v'_y = \frac{-2[(x+1)^2+y^2] - 2y(-2y)}{[(x+1)^2+y^2]^2} = a$$

$$f'(z) = a + ib = \frac{-2(z+1)^2}{(z+1)^4} = \frac{-2}{(z+1)^2}$$

$$f(z) = \frac{2}{z+1} + z_0$$

($z_0$ 为复常数)

9. $$f(z)=\frac{z+1}{z^2-2z} \qquad 0<|z|<2$$

$$f(z)=\frac{A}{z}+\frac{B}{z-2}$$

$$A=\frac{z+1}{z-2}\Big|_{z=0}=-\frac{1}{2} \qquad B=\frac{z+1}{z}\Big|_{z=2}=\frac{3}{2}$$

$$f(z)=\frac{-\frac{1}{2}}{z}+\frac{\frac{3}{2}}{z-2}$$

$$=-\frac{1}{2}z^{-1}-\frac{\frac{3}{4}}{1-\frac{z}{2}}$$

$$=-\frac{1}{2}z^{-1}-\frac{3}{4}\sum_{n=0}^{\infty}\left(\frac{z}{2}\right)^n$$

方法二

$$z^{-1}\left(\frac{z+1}{z-2}\right)$$

$$z^{-1}\left(1+\frac{3}{z-2}\right)$$

$$=z^{-1}\left(1-\frac{\frac{3}{2}}{1-\frac{z}{2}}\right)$$

$$=z^{-1}\left[1-\frac{3}{2}\sum_{n=0}^{\infty}\left(\frac{z}{2}\right)^n\right]$$

$$=z^{-1}-\frac{3}{4}\sum_{n=0}^{\infty}\left(\frac{z}{2}\right)^{n-1}$$

10. $$\oint_{|z|=\frac{1}{2}}\frac{1}{z^{10}(z-1)(z-2)^2}\,dz$$

$$=\oint_{|z|=2}\frac{1}{z^{-10}(z^{-1}-1)(z^{-1}-2)^2}\,z^{-2}\,dz$$

$$=\oint_{|z|=2}\frac{z^{11}}{(1-z)(1-2z)^2}\,dz$$

$$=2\pi i\,\mathrm{Res}\left[\frac{-z^{11}}{(1-2z)^2(z-1)},\,1\right]+2\pi i\,\mathrm{Res}\left[\frac{\frac{1}{4}z^{11}}{(z-\frac{1}{2})^2(1-z)},\,\frac{1}{2}\right]$$

$$=-2\pi i+2\pi i\left(\frac{11}{2^{10}}+\frac{1}{2^{11}}\right)$$

$$\left(\frac{\frac{1}{4}z^{11}}{1-z}\right)'$$

$$=\frac{\frac{1}{4}(11z^{10}+z^{11})}{(1-z)^2}$$

$$=\frac{11}{2^{10}}+\frac{1}{2^{11}}=$$

11. $\int_0^{\frac{\pi}{2}} \frac{2}{a+\sin^2 x}\,dx \quad (a>0)$

$\sin^2 x = \frac{1-\cos 2x}{2}$

$= \int_0^{\frac{\pi}{2}} \frac{2}{a+\frac{1-\cos 2x}{2}}\,dx$

$2x=t$

$dx = \frac{1}{2}\,dt$

$= \int_0^{\frac{\pi}{2}} \frac{4}{2a+1-\cos 2x}\,dx$

$= \int_0^{\pi} \frac{2}{2a+1-\cos t}\,dt$ 偶函数

$z=e^{i\theta}=\cos\theta+i\sin\theta$

$\cos\theta = \frac{z+z^{-1}}{2}$

$= \frac{1}{2}\int_{-\pi}^{\pi} \frac{2}{2a+1-\cos\theta}\,d\theta$

$dz = ie^{i\theta}\,d\theta = iz\,d\theta$

$d\theta = \frac{1}{iz}\,dz$

$= \cancel{\frac{1}{2}} \oint_{|z|=1} \frac{\cancel{2}}{2a+1-\frac{z+z^{-1}}{2}} \frac{1}{iz}\,dz$

$= \frac{1}{i}\oint_{|z|=1} \frac{z}{(4a+2)z-z^2-1}\,dz$

$f(z) = -z^2+(4a+2)z-1$

$\Delta = (4a+2)^2-4 = 16a^2+16a = 16a(a+1) > 0$

$z_2 = \frac{-(4a+2)-4\sqrt{a(a+1)}}{-2} = 2a+1+2\sqrt{a(a+1)}$

$z_1 = 2a+1-2\sqrt{a(a+1)}$

$z_1 \quad z_2$

$z_2 - z_1 = 4\sqrt{a(a+1)} > 4$ 且 $z_2 > 1$

① $z_1 = 1$ $\quad 2a+1-2\sqrt{a(a+1)} = 1$

$a = \sqrt{a(a+1)}$

$\sqrt{a} = \sqrt{a+1}$

② $z_1 = -1$ $\quad 2a+1-2\sqrt{a(a+1)} = -1$

$a+1-\sqrt{a(a+1)} = 0$

$\sqrt{a+1} = \sqrt{a}$

③ $z_1 > 1$ $2a - 2\sqrt{a(a+1)} > 0$ 恒不成立.

$\sqrt{a} > \sqrt{a+1}$ ×

② $-1 < z_1 < 1$ $2a+1 - 2\sqrt{a(a+1)} > -1$ 恒成立.

$\sqrt{a+1} > \sqrt{a}$ √.

$$I = \frac{1}{i} \oint_{|z|=1} \frac{-2}{(z-z_1)(z-z_2)} dz$$

$$= \frac{2\pi i}{i} \operatorname{Res}\left[\frac{-2}{(z-z_1)(z-z_2)}, z_1\right]$$

$$= 2\pi \frac{-2}{z_1 - z_2} = 2\pi \frac{2}{4\sqrt{a(a+1)}} = \frac{\pi}{\sqrt{a(a+1)}}$$